

Subquery 1 — Products costing above the average

```
1 • CREATE DATABASE IF NOT EXISTS StoreDB;
2 • USE StoreDB;
3 • DROP TABLE IF EXISTS Products;
4 • DROP TABLE IF EXISTS Categories;
5 • CREATE TABLE Categories (
6     Category_ID INT PRIMARY KEY,
7     Category_Name VARCHAR(50)
8 );
9 • CREATE TABLE Products (
10     Product_ID INT PRIMARY KEY,
11     Product_Name VARCHAR(50),
12     Category_ID INT,
13     Price INT,
14     Location VARCHAR(50)
15 );
--
16 • INSERT INTO Categories (Category_ID, Category_Name) VALUES
17     (1, 'Electronics'),
18     (2, 'Furniture'),
19     (3, 'Stationery');
20 • INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
21     (201, 'Laptop', 1, 55000, 'Bangalore'),
22     (202, 'Chair', 2, 4500, 'Mysore'),
23     (203, 'Notebook', 3, 120, 'Delhi'),
24     (204, 'Mobile', 1, 30000, 'Mumbai'),
25     (205, 'Table', 2, 8000, 'Chennai');
26 • SELECT Product_ID, Product_Name, Category_ID, Price, Location
27 FROM Products
28 WHERE Price > (SELECT AVG(Price) FROM Products);
```

OUTPUT:

Product_ID	Product_Name	Category_ID	Price	Location
201	Laptop	1	55000	Bangalore
204	Mobile	1	30000	Mumbai
NULL	NULL	NULL	NULL	NULL

Products 1 x

Apply Revert

Outer:

```
1 • CREATE DATABASE IF NOT EXISTS StoreDB;
2 • USE StoreDB;
3 • DROP TABLE IF EXISTS Products;
4 • DROP TABLE IF EXISTS Categories;
5 • CREATE TABLE Categories (
6     Category_ID INT PRIMARY KEY,
7     Category_Name VARCHAR(50)
8 );
9 • CREATE TABLE Products (
10     Product_ID INT PRIMARY KEY,
11     Product_Name VARCHAR(50),
12     Category_ID INT,
13     Price INT,
14     Location VARCHAR(50)
15 );
16 • INSERT INTO Categories (Category_ID, Category_Name) VALUES
17     (1, 'Electronics'),
18     (2, 'Furniture'),
19     (3, 'Stationery');
20 • INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
21     (201, 'Laptop', 1, 55000, 'Bangalore'),
22     (202, 'Chair', 2, 4500, 'Mysore'),
23     (203, 'Notebook', 3, 120, 'Delhi'),
24     (204, 'Mobile', 1, 30000, 'Mumbai'),
25     (205, 'Table', 2, 8000, 'Chennai');
26 • SELECT Product_ID, Product_Name, Category_ID, Price, Location
27 FROM Products
28 WHERE Price >
29     (SELECT AVG(Price) FROM Products);
30
```

OUTPUT:

Result Grid					
Filter Rows:					
Edit:					
Export/Import:					
Wrap Cell Content:					
	Product_ID	Product_Name	Category_ID	Price	Location
▶	201	Laptop	1	55000	Bangalore
	204	Mobile	1	30000	Mumbai
*	NULL	NULL	NULL	NULL	NULL

Products 1 x

Apply Revert

Inner:

```
1 • CREATE DATABASE IF NOT EXISTS StoreDB;
2 • USE StoreDB;
3 • DROP TABLE IF EXISTS Products;
4 • DROP TABLE IF EXISTS Categories;
5 • CREATE TABLE Categories (
6     Category_ID INT PRIMARY KEY,
7     Category_Name VARCHAR(50)
8 );
9 • CREATE TABLE Products (
10     Product_ID INT PRIMARY KEY,
11     Product_Name VARCHAR(50),
12     Category_ID INT,
13     Price INT,
14     Location VARCHAR(50)
15 );

16 • INSERT INTO Categories (Category_ID, Category_Name) VALUES
17     (1, 'Electronics'),
18     (2, 'Furniture'),
19     (3, 'Stationery');
20 • INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
21     (201, 'Laptop', 1, 55000, 'Bangalore'),
22     (202, 'Chair', 2, 4500, 'Mysore'),
23     (203, 'Notebook', 3, 120, 'Delhi'),
24     (204, 'Mobile', 1, 30000, 'Mumbai'),
25     (205, 'Table', 2, 8000, 'Chennai');
26 • SELECT Product_Name, Price
27 FROM Products
28 WHERE Price >
29     (SELECT AVG(Price) FROM Products);
30
```

OUTPUT:

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Product_Name	Price
▶	Laptop	55000
	Mobile	30000

Products 1

×

Result Grid

Read Only

Subquery 2 — Product(s) with the highest price

```
1 • CREATE DATABASE IF NOT EXISTS StoreDB;
2 • USE StoreDB;
3 • DROP TABLE IF EXISTS Products;
4 • DROP TABLE IF EXISTS Categories;
5 • CREATE TABLE Categories (
6     Category_ID INT PRIMARY KEY,
7     Category_Name VARCHAR(50)
8 );
9 • CREATE TABLE Products (
10     Product_ID INT PRIMARY KEY,
11     Product_Name VARCHAR(50),
12     Category_ID INT,
13     Price INT,
14     Location VARCHAR(50)
15 );

16 • INSERT INTO Categories (Category_ID, Category_Name) VALUES
17     (1, 'Electronics'),
18     (2, 'Furniture'),
19     (3, 'Stationery');
20 • INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
21     (201, 'Laptop', 1, 55000, 'Bangalore'),
22     (202, 'Chair', 2, 4500, 'Mysore'),
23     (203, 'Notebook', 3, 120, 'Delhi'),
24     (204, 'Mobile', 1, 30000, 'Mumbai'),
25     (205, 'Table', 2, 8000, 'Chennai');
26 • SELECT Product_ID, Product_Name, Category_ID, Price, Location
27 FROM Products
28 WHERE Price = (SELECT MAX(Price) FROM Products);
```

OUTPUT:

Result Grid					
Filter Rows:					
Edit:					
Export/Import:					
Wrap Cell Content:					
	Product_ID	Product_Name	Category_ID	Price	Location
▶	201	Laptop	1	55000	Bangalore
*	NULL	NULL	NULL	NULL	NULL

Products 2 x Apply Revert

INNER:

```
1 • CREATE DATABASE IF NOT EXISTS StoreDB;
2 • USE StoreDB;
3 • DROP TABLE IF EXISTS Products;
4 • DROP TABLE IF EXISTS Categories;
5 • CREATE TABLE Categories (
6     Category_ID INT PRIMARY KEY,
7     Category_Name VARCHAR(50)
8 );
9 • CREATE TABLE Products (
10     Product_ID INT PRIMARY KEY,
11     Product_Name VARCHAR(50),
12     Category_ID INT,
13     Price INT,
14     Location VARCHAR(50)
15 );

16 • INSERT INTO Categories (Category_ID, Category_Name) VALUES
17     (1, 'Electronics'),
18     (2, 'Furniture'),
19     (3, 'Stationery');
20 • INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
21     (201, 'Laptop', 1, 55000, 'Bangalore'),
22     (202, 'Chair', 2, 4500, 'Mysore'),
23     (203, 'Notebook', 3, 120, 'Delhi'),
24     (204, 'Mobile', 1, 30000, 'Mumbai'),
25     (205, 'Table', 2, 8000, 'Chennai');
26 • SELECT MAX(Price)
27 FROM Products;
```

OUTPUT:

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Result Grid
MAX(Price)				
55000				

Result 1 x Read Only

Subquery 3 — Products costing above Laptop

```
1 • CREATE DATABASE IF NOT EXISTS StoreDB;
2 • USE StoreDB;
3 • DROP TABLE IF EXISTS Products;
4 • DROP TABLE IF EXISTS Categories;
5 • CREATE TABLE Categories (
6     Category_ID INT PRIMARY KEY,
7     Category_Name VARCHAR(50)
8 );
9 • CREATE TABLE Products (
10     Product_ID INT PRIMARY KEY,
11     Product_Name VARCHAR(50),
12     Category_ID INT,
13     Price INT,
14     Location VARCHAR(50)
15 );

16 • INSERT INTO Categories (Category_ID, Category_Name) VALUES
17     (1, 'Electronics'),
18     (2, 'Furniture'),
19     (3, 'Stationery');
20 • INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
21     (201, 'Laptop', 1, 55000, 'Bangalore'),
22     (202, 'Chair', 2, 4500, 'Mysore'),
23     (203, 'Notebook', 3, 120, 'Delhi'),
24     (204, 'Mobile', 1, 30000, 'Mumbai'),
25     (205, 'Table', 2, 8000, 'Chennai');
26 • SELECT Product_ID, Product_Name, Category_ID, Price, Location
27 FROM Products
28 WHERE Price >
29     (SELECT Price FROM Products WHERE Product_Name = 'Table');
30
```

OUTPUT:

Product_ID	Product_Name	Category_ID	Price	Location
201	Laptop	1	55000	Bangalore
204	Mobile	1	30000	Mumbai
NULL	NULL	NULL	NULL	NULL

Subquery 4 — Products in the same category as Laptop

```
1 • CREATE DATABASE IF NOT EXISTS StoreDB;
2 • USE StoreDB;
3 • DROP TABLE IF EXISTS Products;
4 • DROP TABLE IF EXISTS Categories;
5 • CREATE TABLE Categories (
6     Category_ID INT PRIMARY KEY,
7     Category_Name VARCHAR(50)
8 );
9 • CREATE TABLE Products (
10     Product_ID INT PRIMARY KEY,
11     Product_Name VARCHAR(50),
12     Category_ID INT,
13     Price INT,
14     Location VARCHAR(50)
15 );

16 • INSERT INTO Categories (Category_ID, Category_Name) VALUES
17     (1, 'Electronics'),
18     (2, 'Furniture'),
19     (3, 'Stationery');
20 • INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
21     (201, 'Laptop', 1, 55000, 'Bangalore'),
22     (202, 'Chair', 2, 4500, 'Mysore'),
23     (203, 'Notebook', 3, 120, 'Delhi'),
24     (204, 'Mobile', 1, 30000, 'Mumbai'),
25     (205, 'Table', 2, 8000, 'Chennai');
26 • SELECT Product_ID, Product_Name, Category_ID, Price, Location
27 FROM Products
28 WHERE Category_ID =
29     (SELECT Category_ID FROM Products WHERE Product_Name = 'Laptop');
```

OUTPUT:

Result Grid					
Filter Rows:					
Edit:					
Export/Import:					
Wrap Cell Content:					
	Product_ID	Product_Name	Category_ID	Price	Location
▶	201	Laptop	1	55000	Bangalore
	204	Mobile	1	30000	Mumbai
*	NULL	NULL	NULL	NULL	NULL

Products 1 x

Apply Revert

Subquery 5 — Products costing above the average Electronics price

```
1 • CREATE DATABASE IF NOT EXISTS StoreDB;
2 • USE StoreDB;
3 • DROP TABLE IF EXISTS Products;
4 • DROP TABLE IF EXISTS Categories;
5 • CREATE TABLE Categories (
6     Category_ID INT PRIMARY KEY,
7     Category_Name VARCHAR(50)
8 );
9 • CREATE TABLE Products (
10     Product_ID INT PRIMARY KEY,
11     Product_Name VARCHAR(50),
12     Category_ID INT,
13     Price INT,
14     Location VARCHAR(50)
15 );

16 • INSERT INTO Categories (Category_ID, Category_Name) VALUES
17     (1, 'Electronics'),
18     (2, 'Furniture'),
19     (3, 'Stationery');
20 • INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
21     (201, 'Laptop', 1, 55000, 'Bangalore'),
22     (202, 'Chair', 2, 4500, 'Mysore'),
23     (203, 'Notebook', 3, 120, 'Delhi'),
24     (204, 'Mobile', 1, 30000, 'Mumbai'),
25     (205, 'Table', 2, 8000, 'Chennai');
26 • SELECT Product_ID, Product_Name, Category_ID, Price, Location
27 FROM Products
28 WHERE Price >
29     (SELECT AVG(Price) FROM Products WHERE Category_ID = 2);
30
```

OUTPUT:

Result Grid					
Filter Rows:					
Edit:					
Export/Import:					
Wrap Cell Content:					
	Product_ID	Product_Name	Category_ID	Price	Location
▶	201	Laptop	1	55000	Bangalore
	204	Mobile	1	30000	Mumbai
	205	Table	2	8000	Chennai
*	NULL	NULL	NULL	NULL	NULL

Products 3 x Apply Revert

Subquery 6 — Products belonging to a category having a product from Mumbai

```
1 • CREATE DATABASE IF NOT EXISTS StoreDB;
2 • USE StoreDB;
3 • DROP TABLE IF EXISTS Products;
4 • DROP TABLE IF EXISTS Categories;
5 • CREATE TABLE Categories (
6     Category_ID INT PRIMARY KEY,
7     Category_Name VARCHAR(50)
8 );
9 • CREATE TABLE Products (
10     Product_ID INT PRIMARY KEY,
11     Product_Name VARCHAR(50),
12     Category_ID INT,
13     Price INT,
14     Location VARCHAR(50)
15 );

16 • INSERT INTO Categories (Category_ID, Category_Name) VALUES
17     (1, 'Electronics'),
18     (2, 'Furniture'),
19     (3, 'Stationery');
20 • INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
21     (201, 'Laptop', 1, 55000, 'Bangalore'),
22     (202, 'Chair', 2, 4500, 'Mysore'),
23     (203, 'Notebook', 3, 120, 'Delhi'),
24     (204, 'Mobile', 1, 30000, 'Mumbai'),
25     (205, 'Table', 2, 8000, 'Chennai');
26 • SELECT Product_ID, Product_Name, Category_ID, Price, Location
27 FROM Products
28 WHERE Category_ID IN
29     (SELECT Category_ID FROM Products WHERE Location = 'Mumbai');
```

OUTPUT:

Product_ID	Product_Name	Category_ID	Price	Location
201	Laptop	1	55000	Bangalore
204	Mobile	1	30000	Mumbai
* NULL	NULL	NULL	NULL	NULL

Products 1 x Apply Revert

Subquery 7 — Products costing higher than every Furniture product

```
1 • CREATE DATABASE IF NOT EXISTS StoreDB;
2 • USE StoreDB;
3 • DROP TABLE IF EXISTS Products;
4 • DROP TABLE IF EXISTS Categories;
5 • CREATE TABLE Categories (
6     Category_ID INT PRIMARY KEY,
7     Category_Name VARCHAR(50)
8 );
9 • CREATE TABLE Products (
10     Product_ID INT PRIMARY KEY,
11     Product_Name VARCHAR(50),
12     Category_ID INT,
13     Price INT,
14     Location VARCHAR(50)
15 );
```

```
16 • INSERT INTO Categories (Category_ID, Category_Name) VALUES
17     (1, 'Electronics'),
18     (2, 'Furniture'),
19     (3, 'Stationery');
20 • INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
21     (201, 'Laptop', 1, 55000, 'Bangalore'),
22     (202, 'Chair', 2, 4500, 'Mysore'),
23     (203, 'Notebook', 3, 120, 'Delhi'),
24     (204, 'Mobile', 1, 30000, 'Mumbai'),
25     (205, 'Table', 2, 8000, 'Chennai');
26 • SELECT Product_ID, Product_Name, Category_ID, Price, Location
27     FROM Products
28     WHERE Price > (
29         SELECT MAX(Price)
30         FROM Products
31         WHERE Category_ID = 2
32     );
```

OUTPUT:

Product_ID	Product_Name	Category_ID	Price	Location
201	Laptop	1	55000	Bangalore
204	Mobile	1	30000	Mumbai
NULL	NULL	NULL	NULL	NULL

Github:

<https://github.com/manyappoojarybca25-byte/Dbms.git>

